SOFC proof-of-concept · sensor guide

Which sensors tell us what the fuel cell is doing

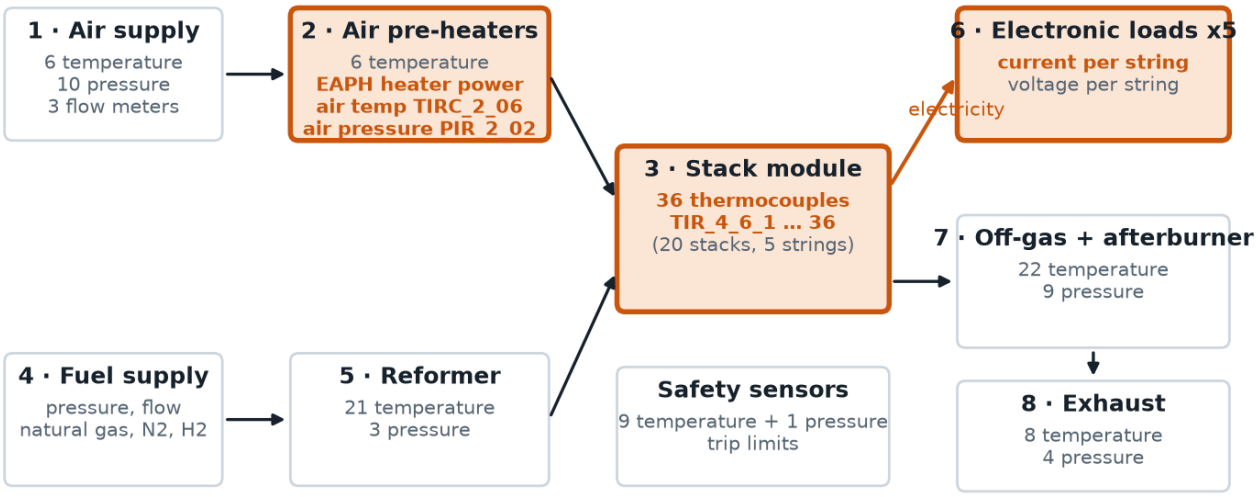
Contents: 1. Which sensors exist · 2. The sensors we use · 3. How they decide the stage

The plant logs **836 data columns**, but only **two measurements decide** which stage it is in: the **temperature inside the stack** and the **electric current it delivers**. Three more signals (heater power, air temperature and air pressure at the stack inlet) mark the exact minute a stage starts or ends.

Why it matters: every later analysis has to compare like with like. A reading taken while the stack warms up means something different from the same reading at full power, so each minute is first labelled with a stage.

1. Where the sensors sit

Air is cleaned and heated (1, 2), natural gas is turned into a hydrogen-rich fuel (4, 5), and both meet in the stack (3), where they make electricity. Electronic loads (6) draw that electricity. Left-over gas is burned (7) and leaves through the exhaust (8).

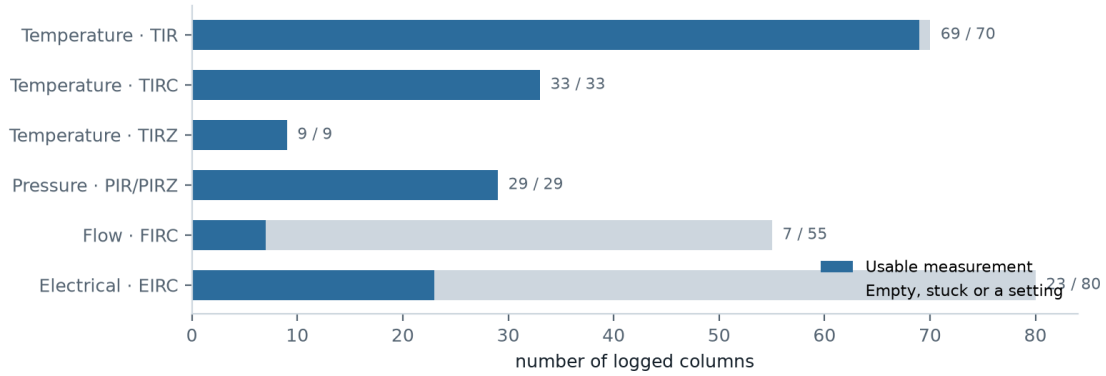


Orange boxes hold the sensors used to name the operating stage.

What kinds of sensors exist

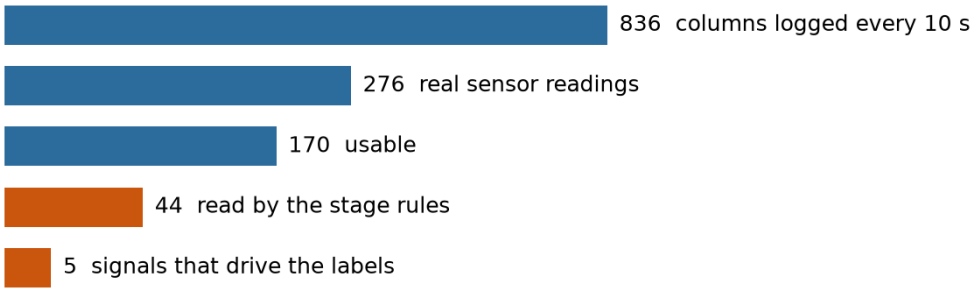
Tag code: first letter = quantity (T temperature, P pressure, F flow, E electrical); I = indicated, R = recorded, C = controlled, Z = safety.

Kind	Tag	What it is	Columns	Usable
Temperature	TIR	Plain thermometers on pipes and in the stack	70	69
Temperature	TIRC	Thermometers a heater controller also reads	33	33
Temperature	TIRZ	Safety thermometers with a trip limit	9	9
Pressure	PIR / PIRZ	Gas pressure in pipes	29	29
Flow	FIRC	Gas flow controllers (air, fuel, N2, H2)	55	7
Electrical	EIRC	Heater power, stack current and voltage	80	23
Total			**276**	**170**



Most flow and electrical columns are settings or on/off switches, not measurements.

From 836 columns to 5 signals

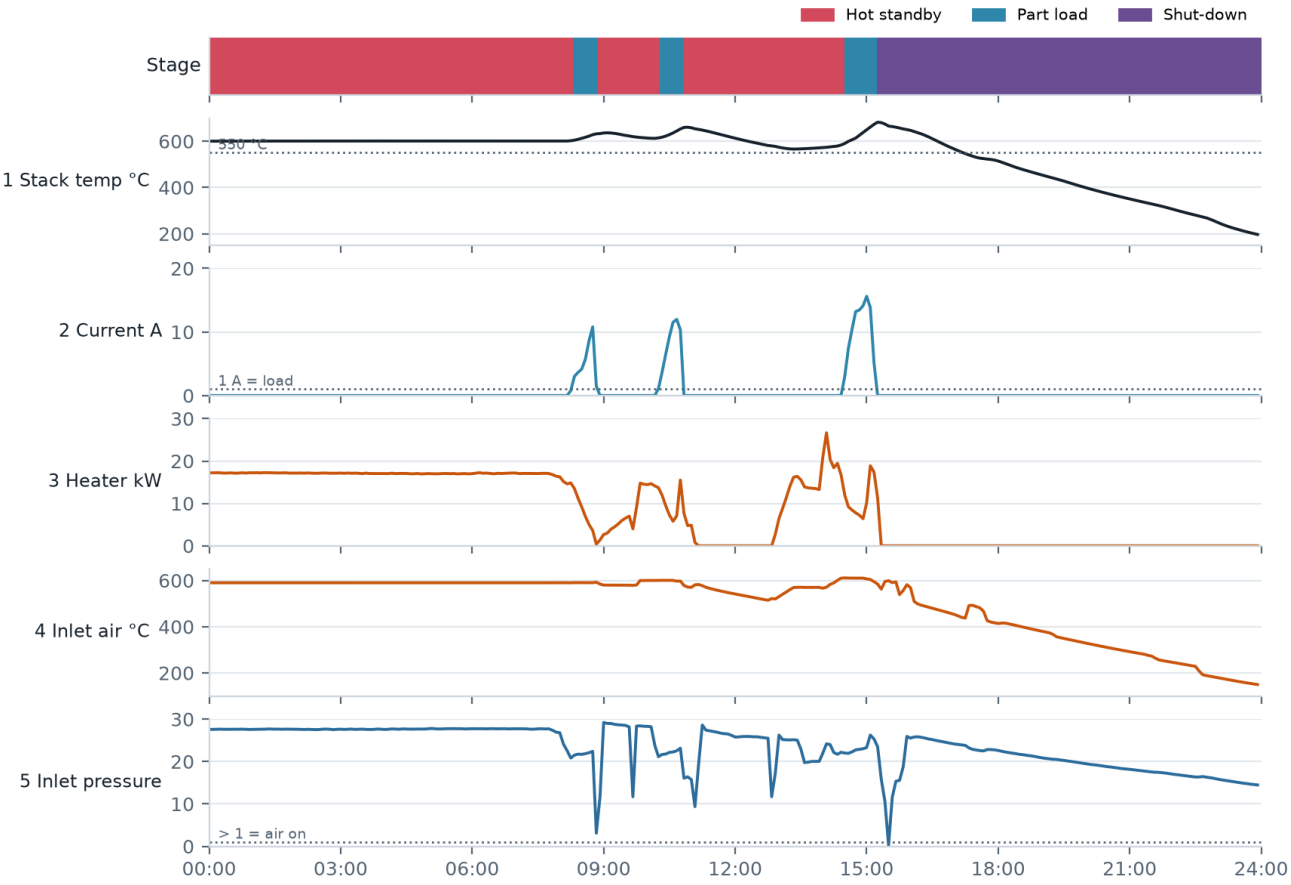


If you remember one thing: Of 836 logged columns, the stage is decided by stack temperature and stack current; three more signals only sharpen the start and end times.

2. The sensors we use

#	Signal	Tag(s)	Role	What it tells us
1	Stack temperature	TIR_4_6_1 … 36 (middle value)	Decides	How hot the stack is and whether it is warming, cooling or steady
2	Stack current	PLC_…_ctEIRC41…45_ rCurrMonitor	Decides	1 A or more for 3 min = load; about 30 A = rated power
3	Air-heater power	EIRC_2_01_ActV_In_ ThreePhasePower	Pins edges	> 0.2 kW = heater on; marks the start of heat-up
4	Air temp at stack inlet	TIRC_2_06	Pins edges	Reacts within minutes of the heater; used to check boundaries
5	Air pressure at stack inlet	PIR_2_02	Pins edges	> 1 = air flowing; separates an air purge from natural cooling

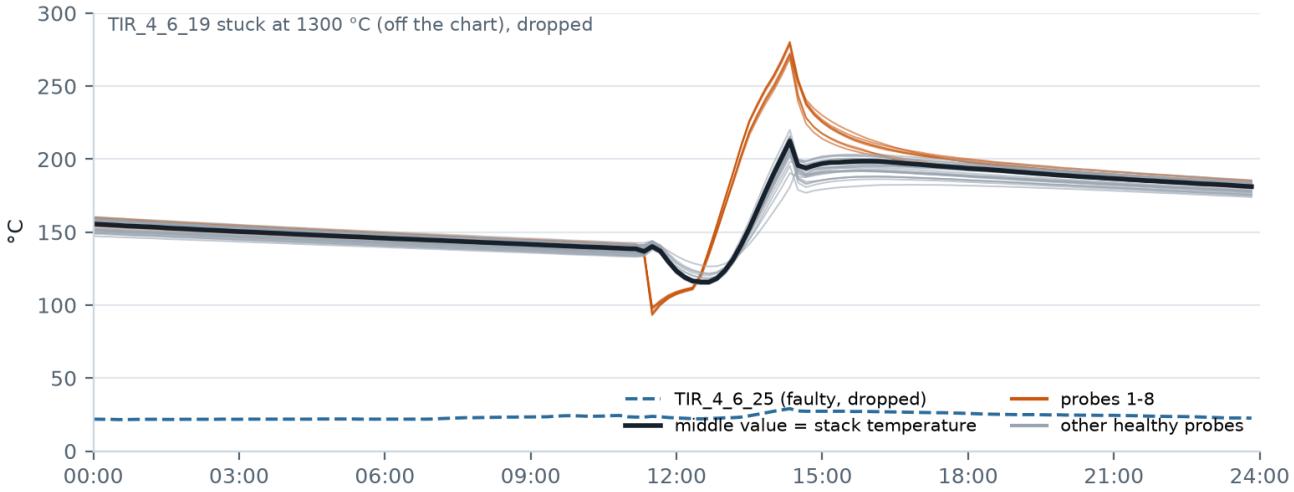
All five signals on a working day (25 March 2026)



Three load periods show as current pulses (2); stack temperature stays above 550 °C between them, so the stage returns to hot standby. Heater power drops during load because the stack heats itself.

Why the 36 probes need cleaning first

Readings outside 0–1000 °C are discarded, and a probe that sits more than 75 °C from the others all day is dropped. Every one of the 38 day files had at least one probe dropped.



All 36 probes on 30 October 2025. TIR_4_6_19 is stuck at 1300 °C; TIR_4_6_25 reads room temperature. Both are dropped.

Sensors deliberately left out

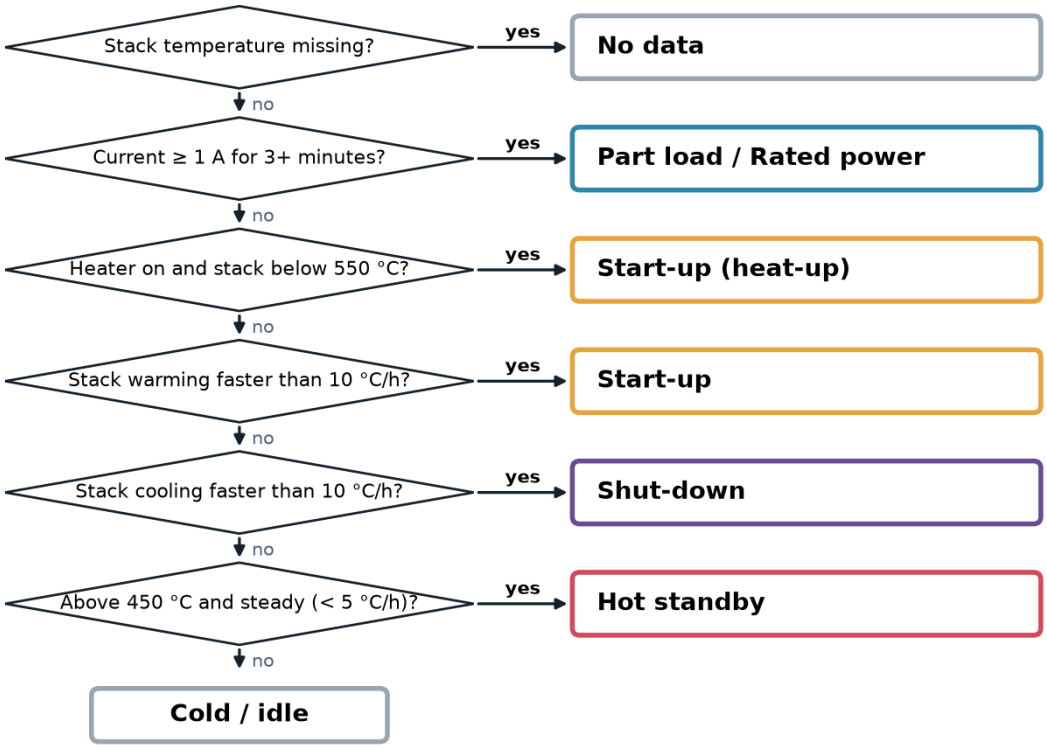
Group	Tags	Why not used
Stack outlet temperatures	TIRC_5_1_x, TIRC_5_2_x	Follow the stack with a delay
Fuel line and reformer	TIR_3_*, TIRC_3_*	Describe fuel processing, not the stack
Exhaust	TIR_7_*, PIR_7_*	Far downstream, slow, mixed with burner heat
Safety sensors	TIRZ_8_*, PIRZ_8_1_2	Limit checks only
Load current, second copy	EIRC_4X_CurrentMon	Garbled on 20 Mar 2026 from about 11:00; the PLC copy is clean
Gas flows	ct_FIRC_*	Not in the current rules; candidate for later

String voltage (rVoltMonitor) is read only to draw stack power on charts; it changes no label.

If you remember one thing: Stack temperature (middle of 36 probes) and stack current name the stage; heater power and inlet air pressure set the minute it changes.

3. How they decide: six questions, asked every minute

The rules ask six questions in a fixed order. The first "yes" gives the stage; if all are "no", the stack is cold or idle.

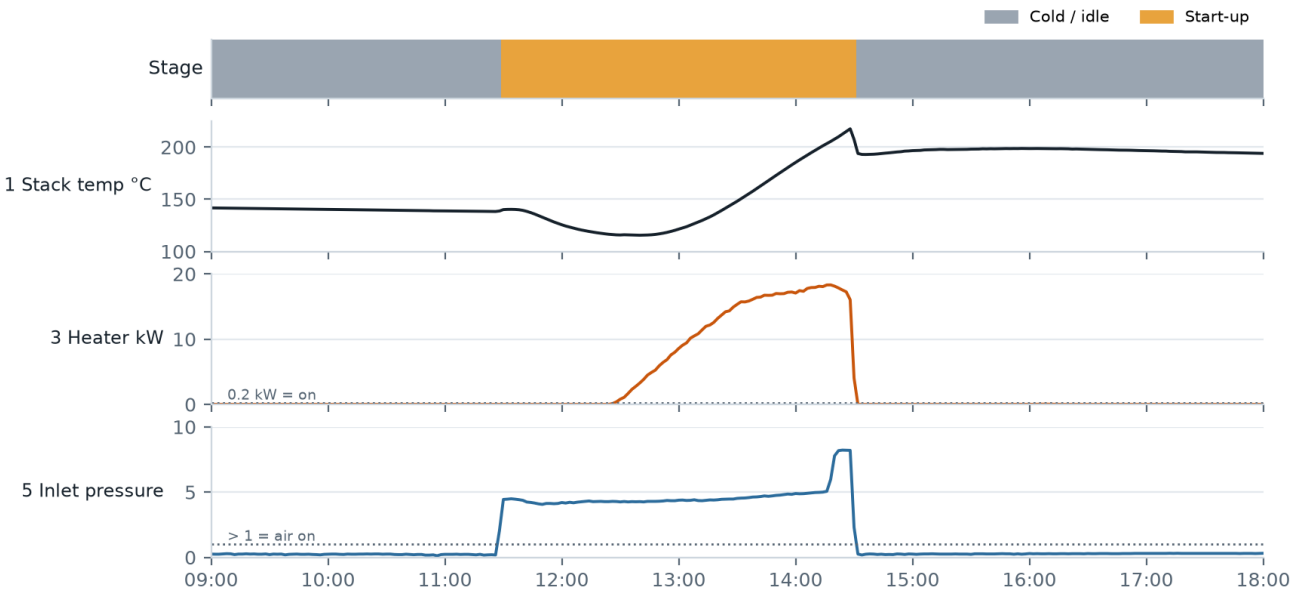


Then the neighbours are checked

A single minute shows	The neighbours reveal	Final label
Cooling, air on	Comes right before a heater heat-up	Start-up · air purge
Cooling	Levels off above 550 °C	Hot standby · cooling back
Warming, no load	Stack was already hot earlier	Hot standby · re-heat
Heater on	After a shut-down, no net warming	Shut-down · controlled cool-down
Current ramping down	Runs straight into a shut-down	Shut-down · load ramp-down
Shut-down block	Between two start-up blocks	Start-up · dip or pause

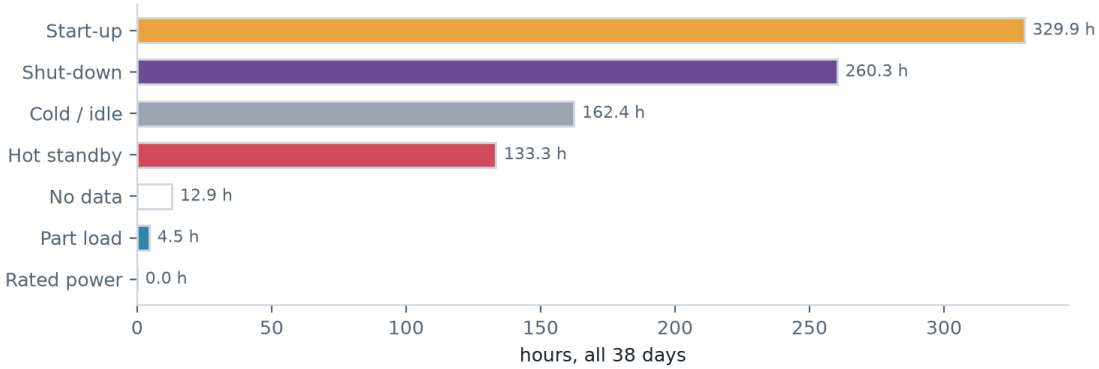
Blocks shorter than 15 minutes merge into their neighbour, never across a load / no-load edge.

A real day: 30 October 2025

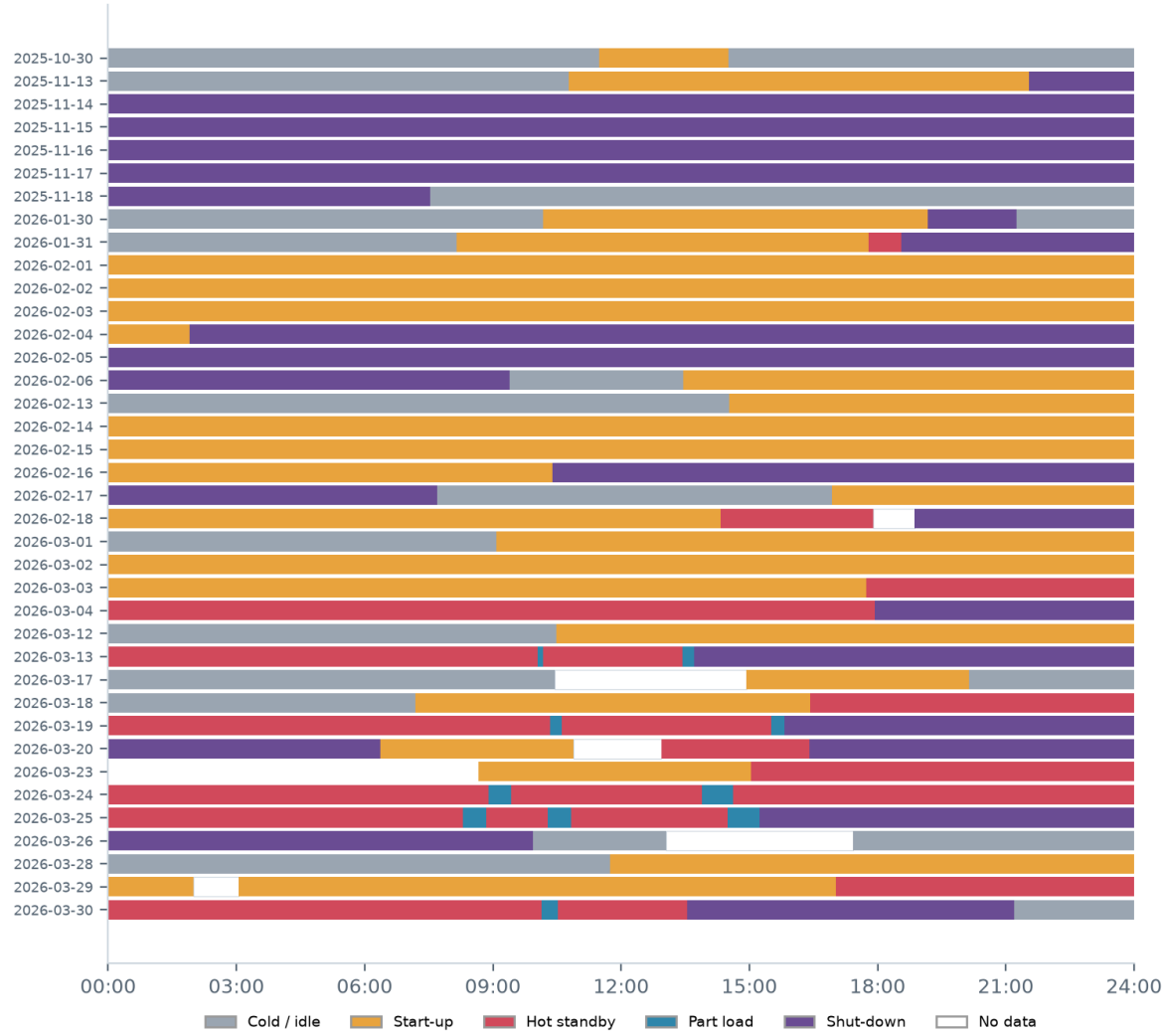


Stage edges fall at 11:29 (air on), 12:27 (heater on) and 14:31 (both off). The stack dips 139 to 116 °C during the purge, climbs to 216 °C, and stops far below hot standby. No current all day.

What the labels found across all 38 days



Current flowed for about 4.5 hours (part load, 5 days); rated power was never reached.



Every day file at a glance, coloured by stage. White = logger gap.

Run it yourself: python data_analysis/operating_states.py 20251030, or open data_analysis/operating_states.ipynb.

If you remember one thing: Current first, then heater, then the stack's temperature trend; the first rule that fits names the stage, and the neighbours correct the obvious mistakes.